

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1395

Greenwood Mo 138 kV
January 24, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1395. The Interconnection Customer proposes a 134.9 MW wind facility interconnecting at the Greenwood Mo 138 kV substation in the Kentucky area. The request was received on January 28, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 360 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$394,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Kirtland Tap - Hopson 100 KV, the Sylvan-Antelope - Hopson 100 KV loads to 105.9% of its 140 MVA rating. Project DFAX: 5.2%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hopon Substation - Sylvan-Antelope 100 KV	Reconductor line	\$1,251,000	32% (\$394,000)
	Total allocated to project		\$394,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.